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Refractory Installation Quality Control Guidelines—Inspection and Testing Monolithic Refractory Linings and Materials

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Manufacturing, Distribution and Marketing Department

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Refractory Installation Quality Control Guidelines—Inspection and Testing Monolithic Refractory Linings and Materials

SECTION 1—GENERAL

1.1 Scope

This document provides guidelines for the installation quality control of monolithic refractory linings and may be used to supplement owner specifications. Materials, equipment, and personnel are qualified by the methods described; and applied refractory quality is closely monitored based on defined procedures and acceptance criteria. The responsibilities of inspection personnel who monitor and control this quality control process are also defined.

1.2 Definition of Terms

For the purposes of this standard, the following definitions apply:

1.2.1 applicator qualification testing: Pre-installation simulation of production work that is sampled and tested as well as visually inspected to verify that application equipment and personnel are capable of meeting specified quality standards.

1.2.2 as-installed testing: Testing of refractory materials sampled from the installation to confirm that they meet specified physical property standards.

1.2.3 contractor: The party or parties responsible for installing refractory in the equipment of the owner.

1.2.4 erosion service: Installations of refractories in fluid solids units, such as transfer and overhead lines, cyclone linings, and deflector shields, in which erosion resistance is a determining feature of lining service life.

1.2.5 heating contractor: Contractor or subcontractor who specializes in the dryout of monolithic refractory linings.

1.2.6 hexalt anchors: Metallic anchor used as an alternative to hexmesh in thin layer, erosion resistant linings; for example, S-Bar, Hexcel, Curl and Tacko anchors, and the like.

1.2.7 independent laboratory: A refractory testing facility not affiliated with the manufacturer or contractor.

1.2.8 inspector: The party or individual whom the owner has contracted or otherwise designated to monitor refractory installation work being conducted by the contractor and supplying material manufacturer(s).

1.2.9 manufacturer: The party or parties supplying the refractory lining materials to be installed in the equipment of the owner.

1.2.10 material qualification testing: Pre-installation testing of refractory materials in which production lots of refractories manufactured for a specific installation are sampled and tested to confirm that they meet specified physical property requirements.

1.2.11 monolithic refractories: Castable or plastic refractories applied by casting, gunning, or hand/ram packing to form monolithic lining structures.

1.2.12 owner: The proprietor of equipment who has engaged one or more parties to install or repair refractory in the equipment.

1.2.13 production run: The quantity of refractory having the same formulation that is prepared in an uninterrupted manufacturing operation.

1.2.14 test sample: That quantity of refractory taken from a single container or installation sequence that is used to make a complete set of test specimens to determine compressive strength, erosion resistance, density, linear change, and/or any other physical property determinations.

1.2.15 test specimen: Individual cubes, bars, or plate test pieces used for physical property testing. Physical property test results for a sample are usually expressed as the average of two or more specimens made up from the same sample.

Additional terms and definitions applicable to this document and the work that it covers are contained as a glossary in Appendix A.

1.3 Quality Control Elements

Key elements of this guideline are given in Table 1 in work chronology identifying responsible parties and objectives. Also indicated are the paragraphs in which detailed requirements for each of the elements are covered.

1.4 Physical Property Requirements

1.4.1 Refractories applied by this guideline shall be sampled and tested to verify that physical properties meet intended criteria. As defined in 2.1.2, product specific physical property requirements shall be determined by agreement prior to material qualification based on sampling/testing procedures described in this document.

1.4.2 The acceptance/rejection criteria for qualification testing is as follows: For both material and applicator

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Table 1—Quality Control: Key Elements

Elements	Actions	Objectives
Documentation (see 2.1)	Owner specification and/or contractor execution plan	Define job specific work scope
Material qualification (see 2.2)	Testing at Independent or manufacturer's laboratory Inspector directs sampling, monitors specimen preparation and witnesses testing[a]	Confirm that materials manufactured for the job are capable of meeting specified physical property standards
Applicator qualification (see 2.3)	Contractor demonstration of capabilities in simulated installation which is witnessed and inspected by the inspector	Confirm that equipment and personnel are capable of installing qualified materials to specified standards
Installation monitoring (see 2.4)	Inspector monitors contractor work and test sample preparation	Confirm that specifications, good practice and installation procedures are maintained
As-installed testing (see 3.1)	Inspector coordinates sampling and testing of as-installed materials	Confirm that installed materials meet specified physical property standards
Pre-dryout inspection	Inspector visual/hammer test inspection of applied linings	Confirm that installed linings meet specification standards
Dryout monitoring (see 4.2)	If dryout by the contractor, the inspector monitors heating rates/hold times	Confirm that agreed on procedure is maintained
Post dryout inspection	Inspector visual/hammer test inspection of applied linings	Confirm that installed linings meet specification standards

Note:
[a]In cases where an independent laboratory is utilized or the contractor assumes complete accountability for as-installed testing results, inspector participation may be waived or reduced by the owner.

qualification testing, the average physical properties for each sample shall fully meet the criteria established for that material in 2.1.2.

1.4.3 Acceptance/rejection criteria for as-installed testing: Based on criteria and procedures agreed to prior to work start, 2.1.2 physical properties criteria shall be extended to account for field conditions as follows in Table 2:

Table 2—Refractories: Physical Properties and Acceptable Results

	Range Of Acceptable Results[a]	
Physical Properties	Minimum[b]	Maximum[b]
Abrasion Loss	none	120%
Cold crushing strength	80%	none
Density	–80 kg/m[c] (–5 lb/ft[c])	+80 kg/m[c] (+5 lb/ft[c])
Permanent linear shrinkage	Zero[c]	120%

Note: kg/m[c] = kilograms per cubic meter; lb/ft[c] = pounds per square foot.
[a]Average of all specimen test results per sample, based on the manufacturer's claimed physical properties for the product tested as reported by a datasheet or other, per 2.1.2.
[b]When the manufacturer claims a range of physical property values for a product, applicable limits shall be the upper and lower limits of that range.
[c]*Zero* means 0.00 percent shrinkage in absolute terms. Products that expand shall not be used unless agreed by the owner.

1.5 Inspection

1.5.1 QUALIFICATIONS OF PERSONNEL

The following qualifications of personnel apply:

a. The inspector shall have no commercial affiliations with the contractor or manufacturer(s), unless otherwise agreed to by the owner.
b. The inspector shall have a working knowledge of applicable standards and terms as defined in attached appendixes of this document.
c. The inspector shall have a working knowledge of requirements defined in this document as well as those provided by the owner for the specific job at hand.

1.5.2 RESPONSIBILITIES

a. The inspector shall monitor qualification and production work conducted by the manufacturer(s) and contractor to ensure compliance with job specifications and agreed-to quality standards.
b. The inspector shall notify the owner and the contractor in a timely manner of any work deficiencies or potential deficiencies based on his or her other monitoring of material qualifications and the installation process.
c. The inspector shall inform the owner of suggestions that, in his or her judgment or experience, will improve the applied lining performance or reduce installation costs. Other than these suggestions, the inspector shall make no engineering decisions contrary or in addition to specified requirements.
d. Any conflict between specified standards and installed refractory quality shall be submitted to the owner for resolution.
e. In cases involving repair and maintenance of existing equipment, the owner may request the inspector to provide

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recommendations concerning the need and extent of repairs and procedures to be used to make those repairs.
f. The inspector shall inspect and hammer test-newly installed linings before dryout and after dryout (when possible), and report any anomalies to the owner.
g. The inspector shall ensure that Material and Applicator Qualification Test results are fully documented and the contractor is provided with all applicable data for the materials that he is installing.
h. The inspector shall check and verify that accurate installation records are being documented by the contractor as per 2.4.8.
i. The inspector shall record all nonconformances and/or potential problems to which the inspector has alerted the contractor and owner.

1.6 References

1.6.1 The following standards are cited in this recommended practice. The latest editions will apply unless otherwise stated.

ASTM[1]

C 16	*Load Subsidence of Refractories*
C 133	*Standard Test Methods for Cold Crushing Strength and Modulus of Rupture of Refractories*
C 181	*Standard Method for Workability Index of Fireclay and High Alumina Plastic Refractories*
C 704	*Abrasion Resistance of Refractory Materials at Room Temperature*
C 1054	*Pressing and Drying Refractory Plastic and Ramming Mix Specimens*

PFI[2]

ES 22	*Recommended Practices for Color Coding of Piping Materials*

1.6.2 Appendix B contains a list of additional standards that the reader may find useful.

SECTION 2—WORK EXECUTION

2.1 Documentation

Supplemental to this document, the owner shall prepare a detailed specification and/or refer to the contractor to prepare a detailed execution plan subject to the owner's approval. These supplemental documents shall be prepared and agreed to in full before work starts. Items covered shall include the following:

2.1.1 DESIGN DETAILS

The following design details apply:

a. Lining products, thickness, method of application, and extent of coverage.
b. Anchor materials, geometry, and layout. Suggested color coding for metallic anchors is given in Appendix C.
c. Surface preparation, welding procedures, and anchor attachment integrity.
d. When used, details of metal fiber reinforcement including dimensions, concentration, type, and metallurgy.

2.1.2 QUALITY STANDARDS:

The following quality standards apply:

a. *Physical property requirements* to be used for qualification and installation quality control by specific product and location where it is being utilized. (These shall be per the manufacturer's datasheet claims unless herein amended.)
b. Sampling procedures as applicable to designation of products to be used in either *erosion service* or *other service* as defined in 2.2.4.
c. Required lining thickness tolerances and criteria for hammer test and allowable cracking in the applied lining.

2.1.3 EXECUTION DETAILS

The following execution details apply:

a. Installation and quality control procedures.
b. Designation of responsible parties.
c. Curing and dryout procedures.

2.2 Material Qualification Testing

2.2.1 All refractories to be installed by gunning, casting, or hand/ram packing shall be tested to comply with specified physical property requirements as determined in 2.1.2. Tested physical properties shall be density, permanent linear change, and cold crushing strength or abrasion resistance per 3.2.

2.2.2 The contractor shall arrange for testing at either an independent laboratory or the manufacturer's plant, and direct the work to assure that mixing techniques, water contents, ambient temperatures, mix temperatures, and so on, adequately represent those needed for production installation. The operative testing party is responsible for conduct of sampling, specimen preparation, testing, and documentation of results.

[1]American Society for Testing and Materials, 100 Bar Harbor Drive, West Conshohocken, PA 19428.
[2]Pipe Fabrication Institute, 612 Lenora Avenue, Springdale, PA 15144.

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2.2.3 The contractor shall inform the owner of testing arrangements and timing so that the owner may notify the inspector to witness or spot check the testing. When engaged as a witness, the inspector shall select the container to be tested and observe all sampling, specimen preparation, and testing. In cases where an independent laboratory is utilized or the contractor assumes complete accountability for as-installed testing results, inspector participation may be waived or reduced by the owner.

2.2.4 Based on the use designation determined by 2.1.2, testing frequency shall be as follows:

a. Erosion service: one sample per 2 tons or less from each production run.
b. Other service: one sample per 5 tons or less from each production run.

2.2.5 As directed by the contractor, test sample refractories shall be mixed and formed using metal or plastic forms at the required specimen dimensions, or larger dimensions, and then cut to the required dimensions after 24-hour cure. For vibration cast installations, vibration may be used in the forming of the test specimens. For pump cast installations, refractory shall be poured into forms. For hand pack installations, refractory shall be hand packed. For gunned installations, refractory shall be gunned or, alternatively, specimens may be hand packed subject to prior approval by the owner of a suitable, product-specific correlation to gunned properties. Plastic and other ramming refractories may be formed using a mallet or handheld pneumatic rammer. Specimen formation using a pneumatic or ramming press, as described by ASTM C-1054, is not permitted.

2.2.6 Every refractory shall be applied within 4 months of initial qualification tests, or 3 months of a succeeding qualification test per the procedures herein. If the initial qualification period is exceeded, the respective batch may be requalified. Requalification permits usage for an additional 3 months after each succeeding qualification test up to the manufacturer's recommended shelf life.

2.2.7 In the event a sample fails to meet specified results, it may be retested. The retest shall be conducted using the same testing facility and inspection, or at a different facility subject to the owner's approval.

2.3 Applicator Qualification Testing

Prior to installation, the contractor shall take the materials qualified for the job, and, using equipment and personnel to be utilized for the installation work, demonstrate that specified quality standards can be met. This shall be done by simulating the installation work, and sampling and testing the applied materials as follows:

2.3.1 PNEUMATIC GUNNING

A test panel shall be prepared by each nozzleman/gun operator team for each refractory being installed with preparation and examination as follows:

a. A test panel shall be fabricated measuring 24 inches × 24 inches (600 millimeters × 600 millimeters) with thickness and anchors the same as the actual installation job.
b. The test panel shall be inclined 45 degrees above the horizontal and supported on a frame so that the panel's midpoint is approximately 6 feet above ground level. The nozzleman/gun operator team shall demonstrate their abilities by gunning the test panel in this inclined position.
c. The test panel shall be sealed and air cured for a period of 18 hours minimum and then sounded for voids and laminations by striking the surface with a ball peen hammer.
d. The test panel shall be constructed with a removable back for visual inspection of the castable. The panel shall also be sectioned and the cut surfaces inspected for voids, laminations, non-uniformities, and rebound entrapment.
e. Test specimens (number and type per 3.2) shall be cut from each panel and tested for compliance to 2.1.2 physical property standards for density, permanent linear change, and cold crushing strength or abrasion resistance.
f. Satisfactory examination and test results per 2.3.1, Items c–e, will serve to qualify the mixing and installation procedures and the nozzleman/gun operator teams. No nozzleman or gun operator shall gun refractory materials until he has been qualified.

2.3.2 CASTING

A mock-up simulating the most difficult piece of the installation work or the size/shape agreed to in the documentation phase (per 2.1.3) shall be completed, and applied materials sampled and tested prior to actual installation work per the following:

a. The mock-up shall simulate forming and placement procedures, installation of refractory around nozzle protrusions, and fit-up tolerances if work involves lining of sections to be fit-up at a later date.
b. For vibration cast installations, the mock-up shall demonstrate the adequacy of vibration equipment and attachment.
c. For pouring and pump cast installations, only vibration that will be used in the actual installation shall be allowed in the mock-up.
d. Test specimens (number and type per 3.2.1) shall be prepared for materials sampled from the mixes prepared for casting in the mock-up and formed in molds using the same level of agitation as the mock-up. Specimens shall be tested per 3.2 and 3.3 for density, permanent linear change, and cold crushing strength or abrasion resistance.
e. Refractory cast in the mock-up shall be cured for 12 hours minimum and then stripped of forms for visual

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inspection only. The applied lining shall be homogeneous and free of segregation and shall meet specified tolerances.
f. Satisfactory examination and test results per 2.3.2, Item d–e, will serve to qualify the mixing and installation procedures as well as mix water levels. The contractor shall not cast refractory linings until he or she has completed a satisfactory mock-up test.

2.3.3 PLACEMENT OF THIN LAYER, EROSION RESISTANT REFRACTORIES

A test panel 12 inches × 12 inches × 3/4 or 1 inch (300 millimeters × 300 millimeters × 20 or 25 millimeters), shall be packed by each applicator and examined as follows:

a. Panel thickness shall be the same as the lining to be installed. Mixing and application techniques, for example, pneumatic ramming, hand packing, orientation sidewall or overhead, and the like, shall also be the same as the installation.
b. Hexmesh or hexalt anchoring system (as the case may be) shall be attached to a backing plate such that the backing plate may be removed and the applied refractory lining examined from the backside after a 24-hour air cure.
c. Test specimens (number and type per 2.3.1) shall be prepared for materials sampled from the mixes applied and formed in molds, using the same placement method as the test panel. Specimens shall be tested per 3.2 and 3.3 for density, permanent linear change, and abrasion resistance.
d. Satisfactory examination and test results per 2.3.3, Items b–c, shall serve to qualify the mixing and installation procedures as well as mix water levels. The contractor shall not apply refractory linings until qualified.

2.3.4 THICK LAYER, PLASTIC INSTALLATIONS

A test panel shall be pneumatically ram packed by each applicator with preparation and examination as follows:

a. The test panel shall be 24 inches × 12 inches (600 millimeters × 300 millimeters) with an applied lining thickness and anchorage that are the same as the actual installation. Anchors shall be attached such that the backing plate may be removed and applied refractory examined from the back side. The backing plate shall be coated with a parting agent to facilitate removal from the applied refractory.
b. Test panel refractory shall be installed by pneumatic ramming in a manner simulating the actual installation (in other words, sidewall or overhead).
c. After refractory installation is completed, the test panel backing plate shall be removed immediately and examined for consolidation and voids.
d. No additional test specimens and testing are required as long as material qualification results and applied refractory workability is satisfactory.
e. Satisfactory results will serve to qualify the equipment, techniques, and applicator. No applicator shall ram pack refractory materials until he or she has been qualified.

2.3.5 CONTRACTOR RESPONSIBILITIES

The following list provides contractor responsibilities:

a. Scheduling of Material Qualification Tests and delivery of those materials and test results to the site.
b. Scheduling and execution of work to qualify all equipment and personnel needed to complete installation work, including documentation and verification by the inspector.
c. Preparation and identification of all testing samples and timely delivery to the testing laboratory.
d. Advance notification to the owner of the time and location where work will take place so that this information can be passed on to the inspector.

2.4 Installation

2.4.1 PACKAGING AND STORAGE

The following applies to packaging and storage:

a. Hydraulic bonded, castable refractories shall be packaged in moisture-proof bags with the product name, batch number, and date of manufacture clearly shown. Bag weight shall be marked, and the weight of refractory in the bag shall not deviate from this value by more that ±2 percent.
b. Chemical setting refractories shall be packaged in heat-sealed plastic to assure vapor-tight enclosure. Mechanical protection shall be provided by cardboard, rigid plastic, or metal outside containers. Each container shall be marked with the production batch number. Packages with broken seals or variation in workability shall be subject to requalification.
c. Refractory materials shall be stored in a weather-protected area. Time limits for Material Qualification Tests (see 2.2.6) shall set refractory shelf life. If the manufacturer's shelf life recommendations are more stringent, the manufacturer's restrictions shall apply.
d. Water used for mixing in the refractory shall be potable.

2.4.2 APPLICATION TEMPERATURE

The following applies to application temperature:

a. The temperature of the air and shell at the installation site shall be between 50°F (10°C) minimum and 90°F (32°C) maximum during refractory installation and the 24 hours thereafter.
b. For cold weather conditions, heating and/or external insulation may be used to maintain temperatures above the minimum requirement.
c. For hot weather conditions, shading, water spraying and/or air conditioning may be used to maintain temperatures below the maximum requirement.

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d. Temperature limits for refractory and mix water shall be in accordance with the manufacturer's requirements.

2.4.3 GUNNING

The following applies to gunning:

a. Pre-wet the refractory by mixing with water prior to charging in the gun to reduce dusting and segregation, while at the same time avoid plugging in the feed hose. Optimum amount, aging, and mixing of the pre-wet shall be determined during Applicator Qualification Tests. Use of pre-wet may be waived with some products subject to the manufacturer's recommendations and the owners approval.
b. Gunning equipment shall provide a smooth and continuous supply of water and material without causing laminations, voids, or rebound entrapment. Shotboards or perpendicular edge cuts shall be used to terminate work areas. When stoppages greater than 30 minutes are encountered or initial set is determined by the inspector, only full thickness lining shall be retained.
c. Start gunning at the lowest elevation, building up the lining thickness gradually over an area of not more than 10 square feet (1 square meter) to full thickness and working in an upward direction to minimize the inclusion of rebound. Rebound material shall not be reused.
d. Downhand gunning beyond 30 degrees below horizontal is prohibited unless agreed to otherwise by the owner. The refractory shall be placed by an alternative placement technique such as casting, hand packing, or repositioning to avoid the downhand orientation.
e. Shotboard height and depth gauges shall be used for thickness measurement guides. After gunning and confirmation of sufficient coverage, the refractory shall be allowed to set (15 minutes to 20 minutes typical) before cutting back with a serrated trowel or currycomb. "Flashing" or interrupted build-up of lining thickness is not permitted after initial set as defined by the surface being exposed for more than 30 minutes or becoming dry to the touch.

2.4.4 CASTING

The following applies to casting:

a. Forming shall be sufficiently strong to support the hydraulic head of wet refractory that it will retain and to resist any mechanical loads, such as vibration. The forms shall be waterproof and leak free. Dimensional tolerances shall meet specified requirements. A release agent such as grease, form release, or wax shall be used to facilitate stripping of the forms.
b. Refractory shall be mixed using procedures, equipment, and water levels demonstrated in Material and Applicator Qualification Tests. For vibration casting in pipe sections, mixer capacity should be sufficient to facilitate placement with no more than 10 minutes between successive mix batches. For pump casting, mixer capacity shall be sufficient to allow for continuous pump operation without stops and starts to wait for material.
c. For vibration casting, two or more rotary vibrators shall be mounted externally on the equipment or component to be lined. Vibrators shall have adequate force to move and consolidate the material being vibrated. Each vibrator shall be independently controlled to focus vibration and prevent segregation due to over vibration. The method of vibrator attachment shall be subject to the owner's approval.
d. For pouring or pump casting, submersion vibrators or rodding may be used to aid refractory flow and filling of the formed enclosure.
e. The newly applied lining shall cure per the manufacturer's recommendation (minimum of 24 hours) before moving the piece or stripping the forms. Curing shall be per 4.1.2.

2.4.5 PLACEMENT OF THIN LAYER, EROSION RESISTANT LININGS

The following applies to thin layer, erosion resistant linings:

a. All air-setting, phosphate bonded refractories shall be mixed in a rotating paddle mixer such as that manufactured by Hobart. The mixer shall have steel paddles and bowls. Aluminum paddles and bowls shall not be permitted due to their potential to react with the acid component in the refractory. Mixing shall be in strict accordance with the manufacturer's recommended procedures using water levels determined during Material and Applicator Qualification Tests.
b. All heat-setting plastic refractories shall be installed at the manufactured consistency. Field water addition or reconditioning is not permitted. Any reconditioning must be performed by the manufacturer under controlled plant conditions, and the reconditioned material fully requalified per 2.2.
c. Refractory shall be applied using a handheld, reciprocating pneumatic rammer, rubber mallet, and/or wood block as demonstrated in Applicator Qualification Tests. During placement, refractory shall be fully compacted in and around the anchor supports to form a homogeneous lining-structure free of voids and laminations.
d. Once consolidated, overfill shall be removed flush with the tops of the hexmesh or hexalt anchors using a trowel or currycomb. The surface shall then be tamped, as necessary, to remove imperfections such as surface tearing and pull away defects.
e. Water slicking of the lining surface is not permitted. Water used to clean and lubricate tools shall be dried off prior to use on the refractory.

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2.4.6 THICK LAYER, PLASTIC INSTALLATIONS

The following applies to thick layer, plastic installations:

a. All heat-setting plastic refractories shall be installed at the manufactured consistency. Field water addition or reconditioning is not permitted. Any reconditioning must be performed by the manufacturer under controlled plant conditions and the reconditioned material fully requalified per 2.2.

b. Refractory shall be removed from the container/plastic wrap only when ready for application. Contents shall be placed on a clean surface for cutting and/or separating pre-cut slices. This work surface shall be maintained to avoid contaminating fresh refractory with dried-out material from previous cutting or separating operations.

c. Refractory shall be ram packed in successive handful-sized clumps using a handheld, reciprocating pneumatic rammer, fully consolidating each clump into a uniform mass and compacting the material in and around the anchor supports to form a homogeneous lining structure free of voids and laminations that is greater than the desired lining thickness.

d. Once placed and consolidated, the lining shall be trimmed to the desired lining thickness using a trowel or currycomb. Cutback material may be reused if workability characteristics are not diminished. Under no circumstances shall dry or crumbly material be installed.

e. The trimmed surface shall then be tamped, as necessary, to remove imperfections such as surface tearing and pull away defects. Water slicking of the lining surface is not permitted.

2.4.7 METAL FIBER REINFORCEMENT

a. Metal fiber reinforcement shall be used only when specified by the owner. Fiber additions shall be uniformly dispersed in the castable without agglomeration.

b. The procedure for adding metal fibers during lining installation shall be as follows:

1. Load castable into mixer and pre-mix.
2. Add pre-wet or mixing water.
3. Using a dispersing device, such as 1/2 inch (13 millimeter) hardware mesh, sieve fibers into castable with mixer operating.

c. Specific details of fiber dimensions, concentration, and metallurgy shall be covered in documentation per 2.1.1. Fiber concentrations typically range 1 to 4 wt percent with effective diameters of 0.010 inch to 0.022 inch (0.3 millimeters to 0.6 millimeters) and lengths 3/4 or 1 inch (19 millimeters or 25 millimeters).

2.4.8 CONTRACTOR RESPONSIBILITIES

a. Advance agreement with the owner on all installation details as defined in 2.1.

b. Execution of installation work, including preparation of as-installed samples per 3.1 for density, permanent linear change, and cold crushing strength or abrasion resistance testing, and timely delivery of those samples to the designated test laboratory.

c. Inspector verified documentation of installation records, including these:

1. Product(s) being applied.
2. Pallet numbers and location where applied.
3. Installation crew members (designating nozzleman/gun operator when gunning).
4. Mixing and/or gunning equipment utilized.
5. Fiber and water percentages.
6. Mixing details including time, temperature, and aging time (if gunned).
7. Location and identity of samples taken for installation quality control.

d. Accountability for installed refractories meeting specified standards, including as-installed testing results as defined in 3.1.4, and lining thickness tolerance limits as defined by 2.1.2.

SECTION 3—TESTING

Unless otherwise agreed per 2.1, the following testing will be performed:

3.1 As-Installed Testing

3.1.1 GUNNING

A minimum of one sample of applied refractory shall be gunned by each gunning crew per material per shift using a "wire mesh basket" which is about 12 inches × 12 inches (300 millimeters × 300 millimeters) and at least 4 inches (100 millimeters) deep. The basket, constructed of 1/2 inch (13 millimeters) wire mesh, shall be supported between anchors on the wall where the lining application is proceeding, filled, and immediately removed. All loose refractory or rebound material shall be removed from the area where the basket was placed during sample preparation. The required test specimens per 3.2 shall be prepared by diamond saw-cutting the specimens from refractory applied in the basket and testing per 3.3.

Alternatively, panels with enclosed sides may be used in place of the wire baskets if the panel dimensions are at least 18 inches × 18 inches × 4 inches (450 millimeters × 450 millimeters × 100 millimeters) and test specimens are cut from the center of the panels to avoid possible rebound traps along the sides of the panels.

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3.1.2 CASTING

A minimum of one sample shall be cast by each mixing crew per material per shift. Samples may be formed directly into test specimens from the refractory being installed or cast into larger forms and cut to the required specimen dimensions after curing. Vibration may be used in casting of samples as applicable to simulate installation work. The required specimens per 3.2.1 shall be tested per 3.3.

3.1.3 PLACEMENT OF THIN LAYER, EROSION RESISTANT LININGS AND PLASTICS

A minimum of one sample shall be packed by each applicator per material per shift. Samples shall be formed directly into test specimens (abrasion plates and linear change bars) from the refractory being installed by the ramming technique used for the installation. The required specimens per 3.2.1 shall be tested per 3.3.

3.1.4 ACCEPTANCE/REJECTION

For each sample, the average physical properties of as-installed tests shall meet the criteria defined in 1.4.3 and the following:

a. Inspector-verified records shall be kept by the contractor to identify samples and the areas of installed lining that they represent.
b. Failure to meet the preceding criteria shall be cause for rejection of the area of the lining that the sample represents.
c. In the event of disagreement over installed refractory quality, core samples may be taken from the questionable area of applied lining and retested using the same criteria.

3.2 Test Specimen Preparation

3.2.1 Based on the use designation determined per 2.1.2, the minimum number of refractory specimens for each sample shall be as follows in Table 3:

3.2.2 Hydraulic bonded, castable refractories shall be cured for a minimum of 24 hours after forming. During this period of time, the refractory shall be covered or sealed with an impermeable material and maintained at an ambient temperature of 70°F to 85°F (20°C to 30°C).

3.2.3 Air-setting, phosphate bonded castable refractories shall be air cured, uncovered for a minimum of 24 hours after forming. During this period of time, the refractory shall be protected from moisture and maintained at an ambient temperature of 70°F to 85°F (20°C to 30°C).

3.2.4 Heat-setting, plastic refractories shall be removed from the forms immediately after forming and allowed to air dry at an ambient temperature of 70°F to 85°F (20°C to 30°C) for a minimum of 24 hours.

Table 3—Test Specimen Preparation: Required Number Per Sample

Type of Test	Number of Specimens	Size of Specimens
FOR EROSION SERVICE		
Abrasion resistance	2	4-1/2 × 4-1/2 × 1 in. (114 × 114 × 25 mm)
Permanent linear change	1	2 × 2 × 9 in. (50 × 50 × 230 mm)
Density	—	(Use abrasion plates and linear change bars)
FOR OTHER SERVICE		
Cold crushing strength	3	2 × 2 × 2 in. (50 × 50 × 50 mm[1])
Permanent linear change	1	2 × 2 × 9 in. (50 × 50 × 230 mm)
Density	—	(Use crushing cubes and linear change bars)

Note: mm = millimeter; in. = inch. Cube loading surfaces shall be parallel to within a tolerance of ±0.8 millimeter (1/32 inch) and 90 degree ±1 degree, whether cast or gunned.

3.2.5 Once refractory specimens have been fully cured and removed from forms and/or cut to required dimensions, they shall be marked with temperature resistant paint (to prevent burn-off during firing), dried, and/or fired as follows:

a. Oven dry (required for heat-setting plastics only): 12 hours minimum at 220°F to 230°F (104°C to 110°C) in a forced air, convection dryer. Heating to this level shall be per manufacturer's recommendations.
b. Oven fire: Heat at 300°F per hour (170°C per hour) to 1500°F per hour (815°C), hold for five hours at 1500°F (815°C); cool at 500°F per hour (280°C per hour) maximum to ambient.

3.3 Test Procedures

3.3.1 COLD CRUSHING STRENGTH

All testing shall be in accordance with ASTM C 133 - 91, and the following:

a. The loading head of the test machine shall have a spherical bearing block.
b. For cast or packed specimens, load shall be applied to either pair of faces cast against the side of the molds. For gunned specimens, load shall be applied perpendicular to the gunning direction, in other words, on cut faces perpendicular to the face of the panel.
c. Bedding material shall be non-corrugated cardboard shims, placed between the test specimen and the loading surfaces. New shims shall be used for each test cube. Shim dimensions shall be approximately 3 inches × 3 inches × 1/16

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inch (75 millimeters × 75 millimeters × 1.5 millimeters) thick, minimum. Two thinner shims making up the same total thickness may be used in place of a single shim.
d. Testing machine minimum sensitivity and maximum loading rate shall be as follows in Table 4:

Table 4—Testing Machine Sensitivity and Loading Rate

Castable Type	Testing Machine Sensitivity[a] N (lb_f)	Testing Machine Loading Rate[b] kPa/sec (psi/min)
> 1600 Kg/m^3 (> 100 lb/ft^3)	2222[c] (500)	290[c] (2500)
960–1600 Kg/m^3 (60–100 lb/ft^3)	444 (100)	35 (300)
< 960 Kg/m^3 (< 60 lb/ft^3)	111 (25)	29 (250)

Note: lb_f = pound-force; kPa/sec = kilopascals per second; psi = pounds per square inch; Kg/m^3 = kilograms per cubic meter; lb/ft^3 = pounds per cubic foot.

[a]If load is registered on a dial, the dial calibration shall permit reading to the nearest load value specified. Readings made within 0.8 millimeter ($^1/_{32}$ inch) along the arc described by the end of the pointer are acceptable.

[b]Loading rate shall be based on the nominal area of the test specimen.

[c]50 percent of the expected load may be applied initially at any convenient rate.

3.3.2 ABRASION (EROSION) RESISTANCE

All testing shall be in accordance with ASTM C 704 - 93 and the following:

a. After firing to 1500°F (815°C), weigh the specimens to the nearest 0.1 gram. Determine the volume of the specimens by measurement of length, width and thickness to the nearest 0.5 millimeter (0.02 inch).
b. Place the 4-$^1/_2$ inches × 4-$^1/_2$ inches (114 millimeters × 114 millimeters) face of the test specimens at a 90 degree angle to the glass nozzle with the troweled surface to be abraded 8 inches (205 millimeters) from the tip of the glass nozzle.
c. Turn on the air pressure and regulate to 448 kilopascals (65 pounds per square inch). Check the air pressure before and after the abrading medium is run through the system.
d. Measure the cabinet pressure using the water manometer, and maintain the pressure in the chamber at 1-$^1/_4$ inches of water (311 pascals) by means of a butterfly valve in the exhaust vent.
e. After the air pressure to the nozzle and the chamber have been adjusted, place 1000 ±5 grams of abrading medium in the reserve funnel. Abrading medium shall be silicon carbide sieve sized to ASTM C 704.
f. Use the silicon carbide only one time before discarding.
g. Remove the refractory specimen from the test chamber, blow off the dust, and weigh to the nearest 0.1 gram.
h. From the initial weight and volume, calculate the bulk density of the specimen to the nearest 0.1 gram.
i. Calculate and report the amount of refractory lost by abrasion in cubic centimeters by determining the weight loss of the abraded sample and dividing the loss by the density of the sample.

3.3.3 DENSITY

Densities shall be determined at room temperature on specimens after firing per 3.2.5, Item b. Testing procedure shall be as follows:

a. Measure specimen dimensions to the nearest 0.02 inch (0.5 millimeters) and weight to the nearest 0.002 pound (1.0 gram).
b. Calculate density by dividing weight by volume and report in units of kilograms per cubic meter or pound per square foot.

3.3.4 PERMANENT LINEAR CHANGE

All testing shall be determined on the 2 × 2 × 9 in. (50 millimeters × 50 millimeters × 230 millimeters) specimens as follows:

a. Freshly placed air-setting and cement-bonded refractories shall be allowed to cure for 24 hours minimum before being removed from the molds or being cut into bar specimens. Chemically bonded, heat-setting plastic refractories shall be removed immediately after forming and carefully placed on a flat, carbon steel plate with a 2 inch × 9 inch (50 millimeter × 230 millimeter) face of the specimen resting on and supported by the expanded lathe during the green measurement and drying operations.
b. At room temperature, measure the length of the specimen along the 9 inch (230 millimeter) dimension to the nearest 0.025 millimeter (0.001 inch) and then oven dry per 3.2.5, Item a. After drying, plastic refractories may be removed from the expanded lathe.
c. When cooled to room temperature, measure the length of the specimen along the 9 inch (230 millimeter) dimension to the nearest 0.001 inch (0.025 millimeter) and then fire per 3.2.5, Item b.
d. When cooled to room temperature, again measure the length of specimen along the 9 inch (230 millimeter) dimension at each of the same four corners of the specimen to the nearest 0.001 inch (0.025 millimeter).
e. Calculate length change for each of the four measurements per specimen and divide by initial green or dried dimension. Report permanent linear change green to fired, and dried to fired as total average percentage shrinkage in length for each specimen to ±0.05 percent.

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SECTION 4—POST INSTALLATION

4.1 Curing of Newly Installed Linings

Newly installed air-setting, hydraulic, and chemically bonded linings shall be allowed to cure at 50°F to 90°F (10°C to 32°C) for 24 hours minimum before initial heating of the lining.

4.1.1 For chemically bonded refractories, the lining surface shall remain uncovered and free from contact with moisture during the curing period.

4.1.2 Sealing and/or excess moisture shall be provided for the curing of hydraulic bonded castables per one of the following options:

a. Applying membrane-type (nonreactive) curing compound to all exposed surfaces before the surface is dry to touch. No part of the lining shall be allowed to air dry more than 2 hours prior to the application of curing compound. The curing compound shall be nonflammable and non toxic, and contain pigmentation that allows for complete visual inspection of coverage. All curing compounds shall be approved by the owner.
b. Wetting the exposed surfaces of the newly installed lining with a fine water spray at approximately 2-hour intervals, such that all surfaces shall be maintained wet to the touch throughout the curing period.
c. Covering the exposed surfaces with polyethylene or a damp cloth within two hours of installation.
d. Having no required coverage on formed surfaces as long as the forms are retained for the full 24 hour curing time.

4.1.3 Heat-setting refractories do not require air curing, but they shall be not be exposed to moisture or freezing conditions prior to initial heating.

4.2 Dryout of Newly Installed Linings

4.2.1 Initial heating of newly installed refractory linings shall be performed by process heating devices or temporary equipment such as electric heating elements or portable burners.

4.2.2 Heating shall be controlled using either process or temporary thermocouples to monitor gas temperatures throughout the newly lined area(s).

4.2.3 Heating rates shall be controlled by thermocouples closest to the heat source. The hold temperatures and durations shall be achieved at all thermocouples including those at gas exits of the newly installed refractory.

4.2.4 The contractor shall review dryout provisions with the manufacturer based on the above criteria and submit a procedure to be approved by the owner.

4.2.5 When cooldown is included in the dryout work scope, cooling rate shall not exceed 1500°F (85°C) per hour.

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APPENDIX A—GLOSSARY

Note: See numbered references at the end of this appendix.

abrasion of refractories [1]: Wearing away of the surfaces of refractory bodies in service by the scouring action of moving solids.

acid-proof brick [2]: Brick having low porosity and permeability, and high resistance to chemical attack or penetration by most commercial acids and some other corrosive chemicals.

acid refractories [3]: Refractories containing a substantial amount of silica, which is reactive with basic refractories, basic slags, or basic fluxes at high temperature.

aggregate [2]: As applied to refractories, a ground mineral material, consisting of particles of various sizes, used with much finer sizes for making formed or monolithic bodies.

air-ramming [2]: A method of forming refractory shapes, furnace hearths, or other furnace parts by means of pneumatic hammers.

air-setting refractories: Compositions of ground refractory materials which develop a strong bond at air ambient temperatures by virtue of chemical reactions within the binder phase that is usually activated by water additions. These refractories include cement and phosphate-bonded castables.

alkali hydralysis: Potentially destructive reactions between unfired hydraulic-setting monolythic refractories, carbon dioxide, alkaline compounds, and water.

alumina [2]: Al_2O_3, the oxide of aluminum; melting point 2050°C (3720°F); in combination with H_2O (water), alumina forms the minerals diaspore, bauxite, and gibbsite; in combination with SiO_2 and H_2O, alumina forms kaolinite and other clay minerals.

alumina-silica refractories [2]: Refractories consisting essentially of alumina and silica, such as high-alumina, fireclay, and kaolin refractories.

alumina-zirconia-silica (AZS): Refractories containing alumina-zirconia-silica as a fusion cast body or as an aggregate used in erosion resistant castables and precast special shapes.

amorphous [2]: Lacking crystalline structure or definite molecular arrangement; without definite external form.

anchor or tieback [4]: Metallic or refractory device that retains the refractory or insulation in place.

applicator qualification testing: A preinstallation simulation of production work that is sampled and tested as well as visually inspected to verify that application equipment and personnel are capable of meeting specified quality standards.

apparent porosity (ASTM C-20) [3]: The relationship of the volume of the open pores in a refractory specimen to its exterior volume, expressed in percentage.

arch: A flat or sloped portion of a fired heater radiant section opposite the floor.

arch brick: A standard brick shape whose thickness tapers along its width.

arch, flat [2]: In furnace construction, a flat structure spanning an opening and supported by abutments at its extremities; the arch is formed of a number of special tapered brick, and the brick assembly is held in place by their keying action. Also called a *jack arch.*

arch, sprung [2]: In furnace construction, a bowed or curved structure that is supported by abutments at the sides or ends only, and which usually spans an opening or space between two walls.

arch, suspended [2]: A furnace roof consisting of brick shapes suspended from overhead supporting members.

asbestos: A group of impure magnesium silicate minerals that occur in a fibrous form.

ash [2]: The noncombustible residue that remains after burning a fuel or other combustible material.

as-installed testing: Testing of refractory materials sampled from the installation to confirm that they meet specified physical property standards.

attrition [2]: Wearing away by friction; abrasion.

basic refractories [3]: Refractories whose major constituent is lime, magnesia, or both, and which may react chemically with acid refractories, acid slags, or acid fluxes at a high temperature.

batch: Quantity of castable refractory produced by a single blending or mixing operation either during production or field mixing.

bauxite [2]: (1) A high-alumina mineral, usually consisting of rounded concretionary grains embedded in clay-like mass, and believed to consist essentially of alumina trihydrate ($Al_2O_3 \cdot 3H_2O$) and alumina hydrate ($Al_2O_3 \cdot H_2O$), in varying proportions. (2) Commercially, bauxite must contain at least 65 percent alumina on a calcined basis.

bend test (of anchors): Inspection technique where metal anchors are physically bent at the weld point to verify the weld integrity to the shell or casing.

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binder [5]: "Cementing" material.

biscuit (of hexmesh lining): A refractory piece formed by a hexmesh cell during lining installation that has a hexagonal shape and the thickness of the hexmesh lining.

bloating: A subsurface defect that can occur in plastic refractory lining systems caused by steam pockets entrapped in the pore structure of the refractory during initial heating due either to rapid heatup or insufficient permeability in the refractory.

breaching section (of furnace): Enclosure in a heat exchanger furnace in which flue gases are collected after the last convection coil for transmission to the stack or outlet ducting.

British thermal unit (BTU) [2]: The amount of heat required to raise the temperature of one pound of water one degree Fahrenheit at standard barometric pressure, and at a standard temperature.

bulk density: The ratio of weight (or mass) to volume in the dried or fired condition.

burn [2]: The degree of heat treatment to which refractory brick are subjected in the firing process; also the degree to which desired physical and chemical changes have been developed in the firing of a refractory material.

burning (firing) of refractories [1]: The final heat treatment in a kiln to which refractory brick are subjected in the process of manufacture, for the purpose of developing bond and other necessary physical and chemical properties.

calcium aluminate cement [3]: The product obtained by pulverizing clinker that consists of hydraulic calcium aluminates formed by fusing or sintering a suitably proportioned mixture of aluminous and calcareous materials.

carbon deposition [2]: The deposition of amorphous carbon, resulting from the decomposition of carbon monoxide gas into carbon dioxide and carbon within a critical temperature range. When deposited within the pores of a refractory, the carbon may build up such pressure that it destroys the bond and causes the refractory to disintegrate.

C-clip (anchors) [2]: A C-shaped metallic anchor used to attach ceramic anchors to the casing or shell of a process unit or fired heater.

castable [1]: A combination of refractory grain and suitable bonding agent that, after the addition of a proper liquid, is generally poured into place to form a refractory shape or structure which becomes rigid because of a chemical action.

casting: The process of placing wet mixed refractory concrete by pouring, pumping, rodding or vibrating.

catalyst [2]: A substance that causes or accelerates a chemical change without being permanently affected by the reaction.

cement [2]: A finely divided substance that is workable when first prepared but becomes hard and stone-like as a result of chemical reaction or crystallization; also, the compact ground mass that surrounds and binds together the larger fragments or particles in sedimentary rocks.

ceramic anchor: Fired refractory device that retains the refractory lining in place.

ceramic bond: The high strength bond that is developed between materials, such as clay and aggregates, as a result of thermochemical reactions which occur when materials are subjected to elevated temperature.

ceramic fiber [4]: Fibrous refractory insulation composed primarily of alumina and silica. Applicable forms include bulk, blanket, paper, module vacuum-formed shape and rope.

ceramics [2]: "Products made of inorganic materials by first shaping them and later hardening them by fire"--F. Singer. Originally, the term ceramics referred only to ware formed from clay and hardened by the action of heat, and to the art of making such ware. However, its significance has gradually been extended by usage, and it is now understood to include all refractory materials, cement, lime plaster, pottery, glass, enamels, glazes, abrasives, electrical insulating products, and thermal insulating products made from clay or from other inorganic nonmetallic mineral substances.

chemical setting [5]: Developing a strong bond by chemical reaction. These refractories include phosphate-bonded plastics and ramming mixes.

chemically-bonded brick [2]: Brick manufactured by processes in which mechanical strength is imparted by chemical bonding agents instead of by firing.

clay [2]: A natural mineral aggregate, consisting essentially of hydrous aluminum silicates (also see FIRECLAY).

cold crushing strength (CCS): A measure of a refractory's ability to resist failure under a compressive load as determined at room temperature after drying or firing. CCS is calculated by dividing the total compressive load by the specimen cross-sectional area.

cold face [3]: The surface of a refractory section not exposed to the source of heat.

compactability [5]: The ease with which the volume of a freshly placed plastic refractory or ramming mix is reduced to a practical minimum, usually by ramming.

congruent melting: The change of a substance, when heated from a solid to a liquid of the same composition (for example, melting of ice to water).

convection [2]: The transfer of heat by the circulation or movement of the heated parts of a liquid or gas.

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convection section (of furnace): The section of a heat exchanger furnace downstream of the radiant section that is closely packed with tubes for optimum convective heat transfer.

conversion (of high alumina cement) [3]: The transformation of the hexagonal metastable hydrates (CAH_{10} or C_2AH_8) to the stable, cubic hydrate (C_3AH_6). The cubic hydrate occupies less volume than the hexagonal hydrates, and this results in an increase in matrix porosity and a possible reduction in concrete strength. (Note: C = CaO, A = Al_2O_3, H = H_2O.)

corbel [2]: A supporting projection of the face of a wall; an arrangement of brick in a wall in which each course projects beyond the one immediately below it to form a support, shelf, or baffle.

corrosion of refractories [1]: Destruction of refractory surfaces by the chemical action of external agencies.

corundum [2]: A natural or synthetic mineral theoretically consisting solely of alumina ($A1_2O_3$). Specific gravity 4.00–4.02. Melting point 3720°F (2050°C). Hardness 8.8.

course [2]: A horizontal layer or row of brick in a structure.

creep [5]: Time-dependent deformation due to sustained load.

cristobalite [2]: A mineral form of silica; stable from 2678°F (1470°C) to the melting point at 3133°F (1723°C). Specific gravity is 2.32. Cristobalite is an important constituent of silica brick.

crown [2]: A furnace roof, especially one which is dome-shaped; the highest point of an arch.

crystal [2]: A chemically homogenous solid body having a definite internal molecular structure, and if developed under favorable conditions, having a characteristic external form, bounded by plane surfaces.

crystalline [2]: Composed of crystals.

curing: Process of bond formation in a newly installed monolithic refractory. For hydraulic bonded castables, this occurs at room temperature and is facilitated by an excess of water being present to react with the cement component. For phosphate bonded plastic refractories, heating to 500°F–700°F (260°C–370°C) is required to form the bond.

cut-back: Pre-set refractory trimmed from the lining surface via a cutting action to the final lining thickness dimension, usually in a gunning installation.

cyclones (of FCCU or fluid coking unit): Components, usually internal, used for the separation of particulate solids from flue or product gas.

density [2]: The mass of a unit volume of a substance. It is usually expressed either in grams per cubic centimeter, or in pounds per cubic foot.

devitrification [2]: The change from a glassy to a crystalline condition.

dryout: The initial heating of a newly installed castable lining in which heating rates and hold times are controlled to safely remove retained water without explosive spalling and to form a well distributed network of shrinkage cracks in the lining.

dual-layer lining: As compared to a "one-shot lining," a refractory lining consisting of two different types of monoliths. Typically, this would consist of a low-density insulating refractory behind a stronger, medium- or high-density refractory.

dusting [2]: Conversion of a refractory material either wholly or in part into fine powder or dust. Dusting usually results from (a) chemical reactions such as hydration; or (b) from mineral inversion accompanied by large and abrupt change in volume, such as the inversion of beta to gamma dicalcium silicate upon cooling.

emissivity, thermal [2]: The capacity of a material for radiating heat; commonly expressed as a fraction of percentage of the ideal "black body" radiation of heat, which is the maximum theoretically possible.

erosion of refractories [2]: Mechanical wearing away of the surfaces of refractory bodies in service by the washing action of moving liquids or gasses, such as molten slags or high-velocity particles.

erosion resistance (as applies to ASTM C-704 test results): Volume of refractory loss, as measured in cubic centimeters, after abrading the surface of a test specimen with 1000 grams of SiC grit in accordance with ASTM C-704. The lower the amount of cubic centimeters loss, the higher the erosion resistance of the refractory.

erosion resistant lining [2]: Single layer of erosion-resistant castable refractory retained in hexmesh or with submerged studs/wires when metal reinforcing is incorporated in the material.

erosion service: Installations of refractories in FCCU components, such as transfer and overhead lines, cyclone linings, and deflector shields, in which erosion resistance is a determining feature of lining service life.

expansion joint [5]: A separation between adjoining parts of a refractory lining which allows small expansive movements, such as those caused by thermal changes.

explosive spalling [3]: A sudden spalling which occurs as the result of a build-up of steam pressure caused by too-rapid heating of a castable refractory on first firing.

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extrude [4]: To force a plastic refractory through a die by the application of pressure.

extrusion [2]: A process in which plastic material is forced through a die by the application of pressure.

field mix [3]: A refractory concrete mix which is designed and formulated at or near a particular job site.

firing [5]: The process of heating refractories to develop desired properties.

firebrick [2]: Refractory brick of any type.

fireclay [2]: An earthy or stony mineral aggregate which has as the essential constituent hydrous silicates of aluminum with or without free silica, plastic when sufficiently pulverized and wetted, rigid when subsequently dried, and of sufficient purity and refractoriness for use in commercial refractory products.

fireclay brick [2]: A refractory brick manufactured substantially or entirely from fireclay.

flash coat: A layer coat of refractory, usually gunned, which is applied over refractory that has already been applied and allowed to set up.

flexmesh: a longitudinally hinged version of hexmesh supplied in flexible rolls for easy access through vessel openings to installation area and ready fit to curved surfaces.

fluid catalytic cracking unit: Also known as FCCU or *Cat Cracker*, a refining process consisting of reactor and regenerator vessels, and interconnecting piping in which particulate catalyst is circulated at elevated temperatures to upgrade low-value feedstock to high-value products such as heating oil, gasoline components, and chemical feedstocks.

flux [2]: A substance or mixture that promotes fusion of a solid material by chemical action.

fluxing [2]: Fusion or melting of substance as a result of chemical action.

flux load (in welding): Addition of an alumina ball to enhance weldability during stud-welding metallic components such as anchors.

footed anchor: Metallic anchor, usually v-stud, which has a foot-shaped configuration at the base to aid weld attachment to the shell.

fractionator (of FCCU or fluid coking unit): Vessel downstream of the reactor used to separate different product fractions.

friable [2]: Easily reduced to a granular or powdery condition.

fused-cast refractories [2]: Refractories formed by electrical fusion followed by casting and annealing.

fused silica: Silica in a fused or vitreous state produced by arc melting of sand. Castables containing fused silica aggregate have low thermal conductivity and low thermal expansion useful in thermal shock applications such as seal pots.

fusion [2]: A state of fluidity or flowing, in consequence of heat; the softening of a solid body, either through heat alone or through heat and the action of a flux, to such a degree that it will no longer support its own weight, but will slump or flow. Also the union or blending of materials, such as metals, with the formation of alloys.

fusion point [2]: The temperature at which melting takes place. Most refractory materials have no definite melting points, but soften gradually over a range of temperatures.

glass [2]: An inorganic product of fusion which has cooled to a rigid condition without crystallizing.

grain size [2]: As applied to ground refractory materials, the relative proportions of particles of different sizes; usually determined by separation into a series of fractions by screening.

green refractory (monolithic linings): A newly installed refractory before it is exposed to dryout or initial heating.

grout [2]: A suspension of mortar material in water, of such consistency that when it is poured upon horizontal courses of brick masonry, it will flow into vertical open joints.

gunning [2]: The application of monolithic refractories by means of air placement guns.

gun operator: Individual in a dry gun operation who controls material charging, flow rate and air flow of the gunning machine.

hammer test (of refractory lining): A subjective test of green or fired refractories in which the lining is impacted with a hammer to gauge soundness and uniformity via audible resonance.

hand packing: Castable installation technique whereby refractory is placed by packing successive handfuls of material to the desired shape. Refractory must be mixed at a consistency that is stiff enough for the placed refractory to hold its shape, while at the same time wet and sticky enough so that the lining formed is structurally homogenous.

heat curing: Process of heating used to develop bonding in refractories such as phosphate bonded refractories. With these refractories, heat curing is concurrent with dryout, but not necessarily interchangeable with use of the term, as *dryout* refers only to the elimination of retained water within the lining system.

heat-setting refractories [2]: Compositions of ground refractory materials which require relatively high tempera-

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tures for the development of an adequate bond, commonly called the *ceramic bond.*

heavy weight castables: Castable refractories with densities roughly greater than 100 pounds per cubic foot (1.6 kilograms per cubic meter).

hexalt anchor: A metallic anchor used as an alternative to hexmesh in thin layer, erosion-resistant linings, for example, S-Bar, Hexcel, Curl, and Tacko anchors, and so forth.

hexmesh: A metallic anchoring system usually 3/4-inch or 1-inch (19 or 25 millimeters) thick that is constructed of metal strips joined together to form hexagonal shaped enclosures where erosion resistant refractory is packed after welding to the base plate steel.

high-alumina refractories [2]: Alumina-silica refractories containing 45 percent or more alumina. The materials used in their production include diaspore, bauxite, gibbsite, kyanite, sillimanite, alusite, and fused alumina (artificial corundum).

high-duty fireclay brick [2]: Fireclay bricks which have a pyrometric cone equivalent (PCE) not lower than Cone 31 1/2 nor above 32 1/2–33.

hot face [3]: The surface of a refractory section exposed to the source of heat.

hydrate/hydration: Chemical reactions between refractory cement components and water that cause the applied lining to develop green strength.

hydraulic-setting refractories [2]: Compositions of ground refractory materials in which some of the components react chemically with water to form a strong hydraulic bond. These refractories are commonly known as *castables.*

incongruent melting: Dissociation of a compound on heating, with the formation of another compound and a liquid each having a different composition from the original compound.

independent laboratory: Refractory testing facility not affiliated with and manufacturer or contractor.

insulating castable [5]: A castable refractory with a relatively low thermal conductivity: it usually has a low in-place density of less than 100 pounds per cubic foot (1.6 kilograms per cubic meter).

insulating firebrick [1]: A refractory brick characterized by low thermal conductivity and low heat capacity.

insulating refractory concrete [3]: Refractory concrete having a low thermal conductivity—it usually has a low density.

inversion [1]: A change in crystal form without change in chemical composition; as for example, the change from low-quartz to high-quartz, or the change from quartz to cristobalite.

isomorphous mixture [2]: A type of solid solution, in which mineral compounds of analogous chemical composition and closely related crystal habit crystallize together in various proportions.

kaolin [2]: A white-burning clay having kaolinite as its chief constituent. The specific gravity is 2.4–2.6. The PCE of most commercial kaolins ranges from Cone 33 to Cone 35.

key [2]: In furnace construction, the uppermost or the closing brick of a curved arch.

key brick: A standard brick shape whose width tapers along its length.

K-factor [2]: The thermal conductivity of a material, expressed in standard units.

lamination defect: A plane of weakness within a monolithic refractory lining that is parallel to the hot face of the lining and permits separation into layers.

lightweight refractory concrete [3]: Refractory concrete having a unit weight less than 100 pounds per cubic foot (1.6 kilograms per cubic meter).

load subsidence: A refractory's load-bearing strength as determined by specimen dimensional changes under a compressive load at high a temperature, per ASTM C 16.

loss on ignition [2]: As applied to chemical analyses, the loss in weight which results from heating a sample of material to a high temperature, after preliminary drying at a temperature just above the boiling point of water. The loss in weight upon drying is called *free moisture*; that which occurs above the boiling point, *loss on ignition.*

low-duty fireclay brick [2]: Fireclay brick which has a PCE not lower than Cone 15, nor higher than 28–29.

material qualification testing: Preinstallation testing of refractory materials in which production lots of refractories manufactured for an installation are sampled and tested to confirm that they meet specified physical property requirements.

matrix [5]: The continuous phase in an emplaced refractory.

medium-duty fireclay brick [2]: A fireclay brick with a PCE value not lower than Cone 29 nor higher than 31–31 1/2.

medium weight castables: Castable refractories with densities roughly between 100 pounds per cubic foot (1.6 kilograms per cubic meter) and 50 pounds per cubic foot (0.8 kilograms per cubic meter).

melting point [2]: The temperature at which crystalline and liquid phases having the same composition coexist in

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equilibrium. Metals and most pure crystalline materials have sharp melting points, in other words, they change abruptly from solid to liquid at definite temperatures (see *congruent melting*). However, most refractory materials have no true melting points, but melt progressively over a relatively wide range of temperatures (see *incongruent melting*).

metal fiber reinforcement: Metal fibers, usually 3/4 inch–1 inch (19 millimeters–25 millimeters) in length, blended into a castable refractory, typically during the mixing operation, at a quantity of up to 1 volume percent of the refractory. Metal fiber reinforcement is used to improve applied lining toughness and shrinkage crack distribution.

mica [2]: A group of rock minerals having nearly perfect cleavage in one direction and consisting of thin elastic plates. The most common varieties are muscovite and biotite.

micron [2]: The one-thousandth part of a millimeter (0.001 millimeters); a unit of measurement used in microscopy and to define the particle size of FCCU catalysts.

mineral [2]: A mineral species is a natural inorganic substance which is either definite in chemical composition and physical characteristics or which varies in these respects within definite natural limits. Most minerals have a definite crystalline structure; a few are amorphous.

modulus of elasticity (physics) [1]: A measure of the elasticity of a solid body; the ratio of stress (force) to strain (deformation) within the elastic limit.

modulus of rupture (MOR)[2]: A measure of the transverse or "cross-breaking" strength of a solid body. MOR is calculated using the total load at which the specimen failed, the span between the supports, and the dimensions of the specimen.

monolithic lining [2]: A castable lining without joints, formed of material which is rammed, cast, gunned, or sintered into place.

monolithic refractories: Castable or plastic refractories applied by casting, gunning, or hand/ram packing to form monolithic lining structures.

mortar (refractory) [1]: A finely ground preparation which becomes plastic and trowelable when mixed with water and is suitable for use in laying and bonding refractory bricks together.

neutral refractories [1]: Refractories that are resistant to chemical attack by both acidic and basic slags, refractories, or fluxes at high temperatures.

nine-inch equivalent [2]: A brick volume equal to that of a standard 9 inches × 4 1/2 inches × 2 1/2 inches straight brick; a unit of measurement of brick quantities in the refractory industry.

normal-weight refractory concrete [3]: Refractory concrete having a unit weight greater than 100 pounds per cubic foot (1.6 kilograms per cubic meter).

nozzleman: Individual at the point of application in a dry gun operation who controls water addition via water valve and material build up via maneuvering and positioning of the outlet nozzle.

off-set lance: Hexmesh manufactured with lance tabs off center.

one-shot lining: A lining composed of a single layer of the one type of castable refractory.

overlay: A layer coat of refractory, usually troweled on, which is applied to an existing lining in an attempt to extend the lining life.

overspray: A cement-rich layer of refractory that deposits on exposed surfaces around a gunning installation site from airborne, wetted refractory dust generated by the gunning operation.

pallet: Quantity of refractory described by amount contained on a shipping pallet.

perlite [1]: A siliceous glassy rock composed of small spheroids, varying in size from small shot to peas; combined with water content, 3 percent to 4 percent. When heated to a suitable temperature, perlite expands to form a lightweight glassy material with a cellular structure.

permanent linear change (PLC): A measure of a refractory's permanent dimensional changes as a result of heating to a specific temperature. A specific specimen dimension is measured before and after heating at room temperature. PLC is calculated by the percentage change in these measurements.

permeability [2]: The property of porous materials which permits the passage of gases and liquids under pressure. The permeability of a body is largely dependent upon the number, size, and shape of the open connecting pores, and is measured by the rate of flow of a standard fluid under definite pressure.

plastic refractory [3]: A moldable refractory material that can be extruded and has a level of workability that permits it to be pounded into place to form a monolithic structure.

plasticity [2]: That property of a material that enables it to be molded into desired forms, which are retained after the pressure of molding has been released.

plenum (of FCCU or fluid coking unit): Enclosure inside the top head of a reactor or regenerator vessel which supports the cyclones and in which gases exiting the cyclone outlets are collected.

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pores [2]: As applied to refractories, the small voids between solid particles. Pores are described as "open" if permeable to fluids; "sealed" if impermeable.

porosity of refractories [2]: The ratio of the volume of the pores or voids in a body to the total volume, usually expressed as a percentage. The "true porosity" is based upon the total pore-volume; the "apparent porosity" upon the open pore-volume only.

pre-wetting (gunning): A technique used with dry gunning machines where a small quantity of water is mixed into the dry refractory before charging into the gun to reduce rebound and dust, and improve wetting of the cement in the gunning operation.

production run: The quantity of refractory having the same formulation that is prepared in an uninterrupted operation of manufacturing.

pump casting: Castable installation technique in which refractory is mixed with water and pumped through piping and/or hoses to a site where it is poured from the outlet nozzle directly into a formed enclosure.

punky: A condition describing a refractory lining that is soft and friable.

pyrometric cone [2]: One of a series of pyramidal-shaped pieces consisting of mineral mixtures and used for measuring time-temperature effect. A standard pyrometric cone is a three-sided truncated pyramid; and, approximately, is either 2 5/8 inches (66 millimeters) high by 5/8 inch (16 millimeters) wide at base or 1 1/8 inches (29 millimeters) high by 3/8 inch (16 millimeters) wide at the base. Each cone is of a definite mineral composition; it bends at a definite temperature.

pyrometric cone equivalent (PCE) [2]: The number of that standard pyrometric cone whose tip would touch the supporting plaque simultaneously with a cone of the refractory material being investigated, when tested in accordance with the method of test for pyrometric cone equivalent (PCE) of refractory materials (see ASTM C24-79).

radiant section (of furnace): The hottest section of a heat exchanger furnace near the burners in which radiant heat transfer is dominant.

ramming mix [3]: A refractory material, usually tempered with water, that cannot be extruded but has suitable properties to permit ramming into place to form a monolithic structure.

reactor (of FCCU or fluid coking unit): The vessel in which cracking reaction occurs or is completed and product gases are separated from coke and/or catalyst particulate. Usually operates at 900°F–1000°F (480°C–540°C).

rebound: Aggregate and/or cement which bounces away from a surface against which refractory is being projected by gunning.

reducing atmosphere [3]: An atmosphere that, at high temperature, lowers the state of oxidation of exposed materials.

refractories [1]: Nonmetallic materials having those chemical and physical properties that make them applicable for structures, or as components of systems, that are exposed to environments above 1000°F (538°C). While their primary function is resistance to high temperature, they are usually called upon to resist other destructive influences also, such as abrasion, pressure, chemical attack, and rapid changes in temperature.

refractoriness [2]: In ceramics, the property of resistance to melting, softening, or deformation at high temperatures. For fireclay and some high-alumina materials, the most commonly used index of refractoriness is that known as the *pyrometric cone equivalent.*

refractory, (adj.) [2]: Chemically and physically stable at high temperatures.

refractory aggregate [3]: Materials having refractory properties which form a refractory body when bonded into a conglomerate mass by a matrix.

refractory concrete [2]: Concrete which is suitable for use at high temperatures and contains hydraulic cement as the binding agent.

regenerator (of FCCU): Vessel in which coke and residual hydrocarbons are burned off the catalyst and the flue gas is then separated from the catalyst. Usually operates at 1200°F–1400°F (650°C–760°C).

rise of arches [2]: The vertical distance between the level of the spring lines and the highest point of the under surface of an arch.

riser (of FCCU or fluid coking unit): Section of transfer line in which flow is in an upward direction.

sample (for testing): That quantity of refractory taken from a single container or installation sequence that is used to make a complete set of test specimens to determine compressive strength, erosion resistance, linear change and/or any other physical property determinations.

screen analysis [2]: The size distribution of noncohering particles as determined by screening through a series of standard screens.

secondary expansion [2]: The property exhibited by some fireclay and high-alumina refractories of developing permanent expansion at temperatures within their useful range; not the same as overfiring.

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semi-silica fireclay brick [2]: A fireclay brick containing not less than 72 percent silica.

setting [5]: The hardening of a refractory that occurs with time and/or temperature.

sheeting [5]: Spalling of layers from the hot face of a refractory lining.

shelf life [5]: Maximum time interval during which a material may be stored and remain in a usable condition.

shotboard: Temporary containments used in gunning which are set up and secured against anchors to provide a firm surface on which to make perpendicular cold joints at the termination of work areas.

shotcrete [3]: Mortar or concrete projected at high velocity onto a surface; also known as *air blown mortar, pneumatically applied mortar or concrete, blastcrete, sprayed mortar* and *gunned concrete.*

silica [2]: SiO_2, the oxide of silicon. Quartz and chalcedony are common silica materials; quartzite, sandstone and sand are composed largely of free silica in the form of quartz.

single-layer lining: One layer of refractory with or without an anchoring system.

sintering [2]: A heat treatment which causes adjacent particles of material to cohere, at a temperature below that of complete melting.

slag [2]: A substance formed in any one of several ways by chemical action and fusion at furnace operating temperatures:

a. In smelting operations, through the combination of a flux, such as limestone, with the gangue or waste portion of the ore.
b. In the refining of metals, by substances such as lime added for the purpose of affecting or aiding the refining.
c. By chemical reaction between refractories and fluxing agents such as coal ash, or between two different types of refractories.

slagging of refractories [1]: Destructive chemical reaction between refractories and external agencies at high temperatures, resulting in the formation of a liquid.

slumping: Condition of pre-set refractory in which gravitational forces cause it to lose its desired shape.

spalling of refractories [2]: The loss of fragments (spalls) from the face of a refractory structure, through cracking and rupture, with exposure of inner portions of the original refractory mass.

specific gravity [2]: The ratio between the weight of a unit volume of a substance and that of some other standard substance, under standard conditions of temperature and pressure. For solids and liquids, the specific gravity is based upon water as the standard.

The "true specific gravity" of a body is based on the volume of solid material, excluding all pores. The bulk or volume-specific gravity is based upon the volume as a whole, that is, the solid material with all included pores. The apparent specific gravity is based upon the volume of the solid material plus the volume of the sealed pores.

specific heat [2]: The quantity of heat required to raise the temperature of a unit mass of a substance one degree.

specimen (for testing): Individual cube, plate, or bar test pieces used for physical-property testing. Physical property test results for a sample are usually expressed as the average or mean of two or more specimens made up from the same sample.

sprung arch [2]: An arch which is supported by abutments at the side or ends only.

Standpipe (of FCCU or fluid coking unit): Section of transfer line in which flow is in a downward direction.

stud weld (of Anchors): Welding method utilizing an arc- welding machine in conjunction with a timer and a gun.

superduty fireclay brick [2]: Fireclay brick which have a PCE not lower than Cone 33, and which meet certain other requirements as outlined in ASTM C27-70.

suspended arch [2]: An arch in which the brick shapes are suspended from overhead supporting members.

termination strip: Steel bar or ring that is attached to the edge of hexmesh at terminations and sharp bends to retain refractory in partial hexmesh cells.

thermal conductivity [2]: The property of matter by virtue of which heat energy is transmitted through particles in contact.

thermal expansion [2]: The increase in linear dimensions and volume which occurs when materials are heated and which is counterbalanced by contraction of equal amount when the materials are cooled.

thermal shock [3]: The exposure of a material or body to a rapid change in temperature which may have deleterious effect.

thermal spalling [3]: Spalling which occurs as the result of stresses caused by non-uniform heating and/or cooling.

tolerance [2]: The permissible deviation in a dimension or property of a material from an established standard, or from an average value.

transfer line (of FCCU or fluid coking unit): Refractory lined pipe used for the transport of hot particulate medium and gases between process vessels.

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vibration casting: Castable installation technique whereby refractory is mixed with water and placed in a formed enclosure with the aid of vibration which causes the refractory to become "fluid like" and thereby flow and consolidate to the desired shape of the formed enclosure.

vitrification [2]: A process of permanent chemical and physical change at high temperatures in a ceramic body, such as fireclay, with the development of a substantial proportion of glass.

V-anchor: Metallic anchor made from rod or bar stock that is configured in a *V*-shape.

warpage [2]: The deviation of the surface of a refractory shape from that intended, caused by bending or bowing during manufacture.

wedge brick: A standard brick shape whose thickness tapers along its length.

wetting [2]: The adherence of a film of liquid to the surface of a solid.

workability index: A measure of the moldability of plastic refractories as determined in accordance with ASTM C 181. Workability index is commonly used to control consistency of plastic refractories during manufacture and serves as a measure of the facility with which it is rammed, gunned, or vibrated in place.

Y-anchor: Metallic anchor made from rod or bar stock that is configured in a *Y*-shape, usually used for dual layer linings.

Young's modulus [2]: In mechanics, the ratio of tensile stress to elongation within the elastic limit; the modulus of elasticity.

References

1. American Society for Testing and Materials (ASTM), *Standard Definition* C71-88, Conshohocken, PA.
2. *Harbison-Walker Handbook of Refractory Practices,* 1st Edition, 1992.
3. *Refractory Concrete* ACI 547-79, American Concrete Institute, Detroit, MI, 1979.
4. API Standard 560, *Fired Heaters For General Refinery Service,* American Petroleum Institute, Washington, D.C., January 1986.
5. *Refractory Plastics and Ramming Mixes* ACI 547.1R-89, American Concrete Institute, Detroit, MI.

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APPENDIX B—APPLICABLE CODES AND STANDARDS

ACI[1]

547.1R-79 *Refractory Concrete*

547.1R-89 *Refractory Plastics and Ramming Mixes*

ASME[2]

Boiler and Pressure Vessel Code,

Section I *Power Boilers*

Section VIII *Pressure Vessels, Division 1*

ASTM[3]

A167 *Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip*

A176 *Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip*

A569 *Steel, Sheet, and Strip, Carbon (0.15 maximum percent), Hot Rolled, Commercial Quality*

A576 *Steel Bars, Carbon, Hot-Wrought, Special Quality*

A580 *Stainless and Heat-Resisting Steel Wire*

A743/A743M *Iron-Chromium, Iron-Chromium-Nickel, Nickel Base, Corrosion-Resistant for General Applications*

C71 *Standard Terminology Relating to Refractory*

C179 *Standard Test Method for Drying and Firing Linear Change of Refractory Plastic and Ramming Mix Specimens*

C109 *Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or 50-mm Cube Specimens)*

C113 *Standard Test Method for Reheat Change of Refractory Brick*

C134 *Standard Test Method for Size and Bulk Density of Refractory Brick and IFB*

C201 *Standard Test Method for Thermal Conductivity of Refractories*

C401 *Standard Classification of Alumina and Alumina-Silicate Castable Refractories*

C417 *Standard Test Method for Thermal Conductivity of Unfired Monolithic Refractories*

C573 *Standard Methods for Chemical Analysis of Fireclay and High-Alumina Refractories*

C673 *Standard Methods for Chemical Analysis of Fireclay and High-Alumina Plastic Refractories and Ramming Mixes*

C704 *Abrasion Resistance of Refractory Materials at Room Temperature*

C832 *Standard Test Method of Measuring the Thermal Expansion and Creep of Refractories Under Load*

C860 *Determining and Measuring Consistency of Refractory Concretes*

C865 *Firing Refractory Concrete Specimens*

C914 *Standard Test Method for Bulk Density and Volume of Solid Refractories by Wax Immersion*

C1095 REV A *Standard Practice for Calculating Precision Data on Refractories (C08) from Interlaboratory Test Results*

C1113 *Standard Test Method for Thermal Conductivity of Refractories by Hot Wire*

C1171 *Standard Test Method for Quantitatively Measuring the Effect of Thermal Cycling on Refractories*

SSPC[4]

SP3 *Power Tool Cleaning*

[1]American Concrete Institute, 22400 West Seven Mile Road, P.O. Box 19150, Detroit, MI 48219.

[2]American Society for Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

[3]American Society for Testing and Materials, 100 Bar Harbor Drive, West Conshohocken, PA 19428.

[4]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburg, PA 15213–2683.

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APPENDIX C—COLOR CODING FOR METALLIC ANCHORS

1.0 Scope

1.1 The purpose of this color coding guideline is to provide a method for visual identification of metallic anchors used in refractory linings by a general alloy classification.

1.2 Only anchors made of rods and castings, such as V, Steerhorn, and other variations of these that are used in monolithic castable and plastic refractory linings are covered in color codification. Hexsteel and hexalt anchors used in thin erosion-resistant linings and various designs of stud and washer anchors for ceramic fiber linings are not included in this specification as they can be best identified by stamping.

1.3 Identification by this method is not a substitute for PMI (positive material identification) or other permanent manufacturer's markings or labeling that may be required by any owners/users. Also it is not intended that the color coding will be resistant to fading to all conditions including exposure to high-temperature operations. The principal purpose is to ensure identification of the alloy of the anchor to facilitate proper installation, inspection, and storage for future usage.

1.4 Many people are lacking color discrimination. Users of this specification shall therefore ensure that personnel involved in color identification are able to discriminate colors.

2.0 Marking Material Requirements

2.1 Paint shall be the material of marking for color codes. It is specified for efficiency and cost effectiveness. Other marking materials, such as dye, ink, and colored labels, are permissible provided the marking material meets the durability and identification requirements of Section 2.2.

2.2 Paints used for identification markings shall be durable and of identifying colors. Markings of the color-coded anchors shall not fade when stored indoors in a standard warehouse.

2.3 Paint material shall be acrylic, alkyd modified acrylic, or alkyd enamels capable of fast drying.

2.4 Paint shall be free of cadmium, chromium, and lead. Also, the paint shall not contain copper, tin, zinc, chloride, sulfide, and other undesirable elements in any significant quantities.

2.5 The marking requirements described in this color codification shall be applied by the manufacturer (supplier or vendor as applicable) and are supplemental to any other marking and labeling standards and specifications under which the materials may be manufactured.

2.6 Paints contain solvents that pose safety and health hazards. Therefore, paint shall be applied in fire protected, well ventilated areas with proper respirators worn.

2.7 The surface to be color coded shall be cleaned and free of dirt, loose scale, and oil.

3.0 Color Codes Requirements

3.1 Marking for color coding shall be done by painting one or more stripes in top part of both legs of the anchor per dimension descriptions of Table 5.

3.2 For two-component anchors, such as a V screwed or welded on a stud, each part of the anchor shall be color coded per alloy of the part. The limitation of Table 5 shall apply only to the stud part.

3.3 Each stripe representing a color code shall be of single solid color. The color codes shall be per Table 6 as per the Pipe Fabrication Institute Standard ES 22.

Table 5—Dimension of Color Stripe

Color Stripe	Anchor Size 50 mm (2") or Less	Anchor Size 50 mm (2") or More
Single stripe	Full top half of the anchor	25–50 mm (1"-2") wide in top half of the anchor
Double Stripe	2 equal size bands with 3-mm (1/8") minimum gap between bands at least 13 mm (1/2") above the bottom	2 equal-size bands with 6-mm (1/4") minimum gap between bands in top half of the anchor

Note: mm = millimeters; " = inch.

Table 6—Color Codes
(Per Pipe Fabrication Institute ES 22)

Alloy Material	Color Of Stripe[a]
Carbon Steel	None
Carbon steel, killed	2 solid green
304	1 solid black
304L	2 solid black
309	1 solid black, 1 solid brown
310	1 solid green, 1 solid orange
316	1 solid gray
316L	2 solid gray
321	1 solid pink
347	1 solid brown
405	1 solid green, 1 solid black
410S	1 solid green, 1 solid brown
Inconel 600	2 solid blue
Incoloy 800	1 solid black, 1 solid orange

[a]Each marking of color code is a single solid color stripe unless noted otherwise.

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